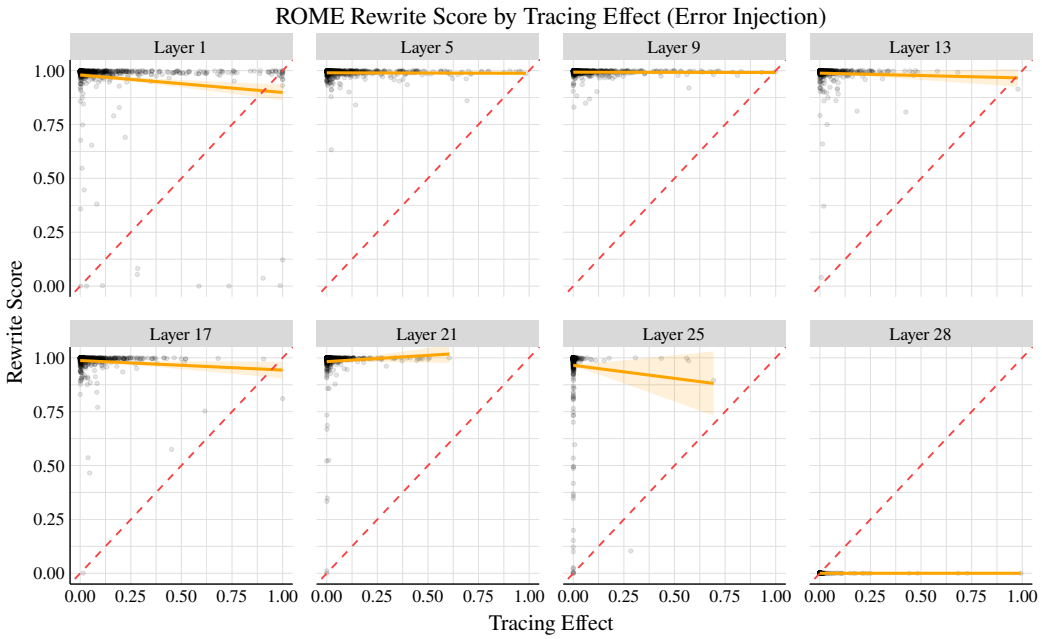


Figure 15: The relationship between ROME edit success and the tracing effect is near zero at most edit layers in the model (for the standard Error Injection editing problem). Red lines show perfect relationships between tracing effect and edit success.

Edit Metric	Regression Metric	Predictor(s)	Value
Rewrite Score	R^2	Layer	0.795
		Tracing Effect	0.042
	RMSE	Layer	0.158
		Tracing Effect	0.341
	MAE	Layer	0.072
		Tracing Effect	0.254
Overall Score	R^2	Layer	0.654
		Tracing Effect	0.059
	RMSE	Layer	0.136
		Tracing Effect	0.223
	MAE	Layer	0.097
		Tracing Effect	0.188

Table 8: **ZSRE regression results** lead us to the same conclusion as our experiments on CounterFact, using ROME editing. RMSE is root mean squared error, and MAE is mean absolute error. Regressions predicting rewrite score (or overall score) from the choice of edit layer achieve much lower prediction errors than regressions using the tracing effect, suggesting that the choice of edit layer is much more important for edit success than the tracing effect.

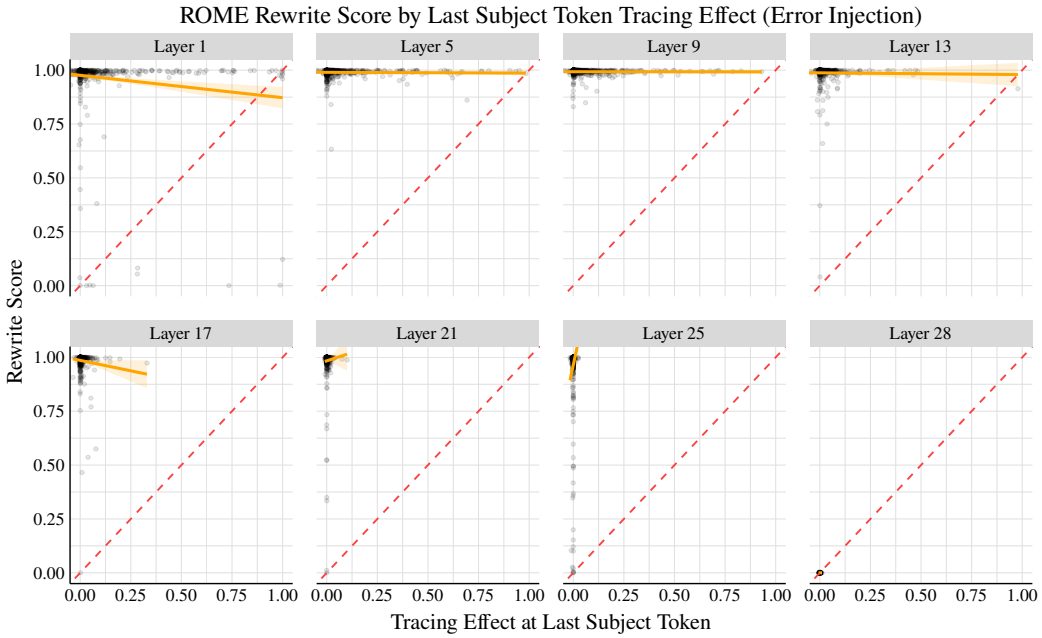


Figure 16: The relationship between ROME edit success and the tracing effect at the last subject token. The ROME method edits a fact by changing the output representation for the MLP layer specifically at the token index corresponding to the last subject token. However, editing performance and tracing effect at this position still do not positively correlate. Note the distribution of points along the x axis changes depending on the choice of edit layer since the distribution of tracing effects is calculated from tracing effects *at that layer*.

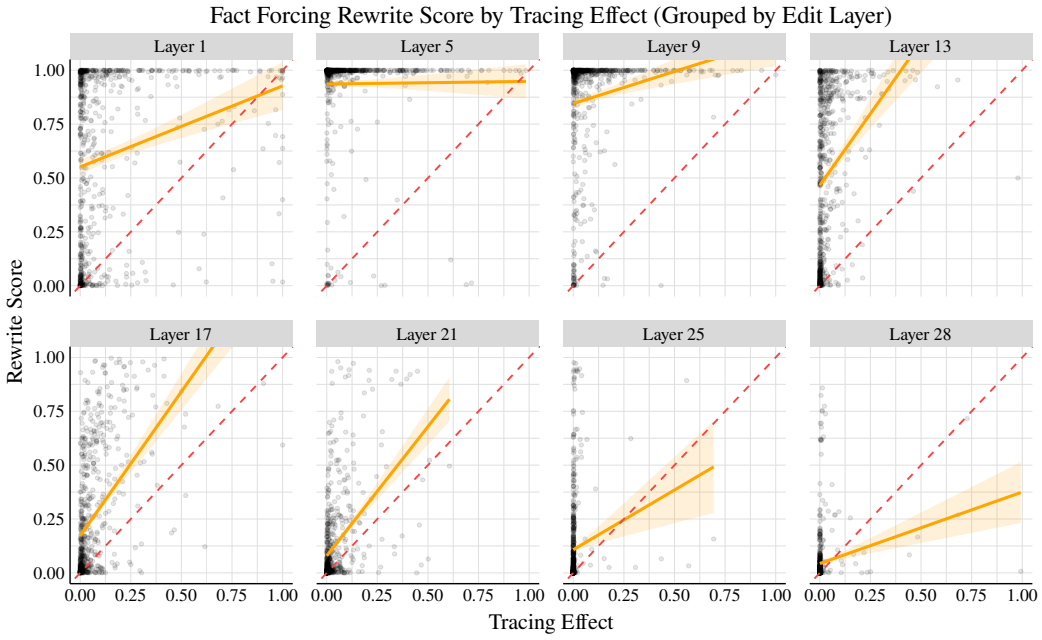


Figure 17: The relationship between Fact Forcing edit success and the tracing effect for constrained finetuning of 5 adjacent layers. "Layer ℓ" indicates the center of this 5-layer interval, and the dashed red lines show a hypothetical perfect relationship between tracing effect and edit success. For many layers, there is a noticeable positive relationship between tracing effects and editing success. Yet, (1) there is a high amount of variance in the outcome, and (2) this variance is largely explained by the edit layer. As a result, tracing effects provide little extra information for predicting edit success beyond the choice of edit layer (about 3% more explained variance; see Fig. 6).

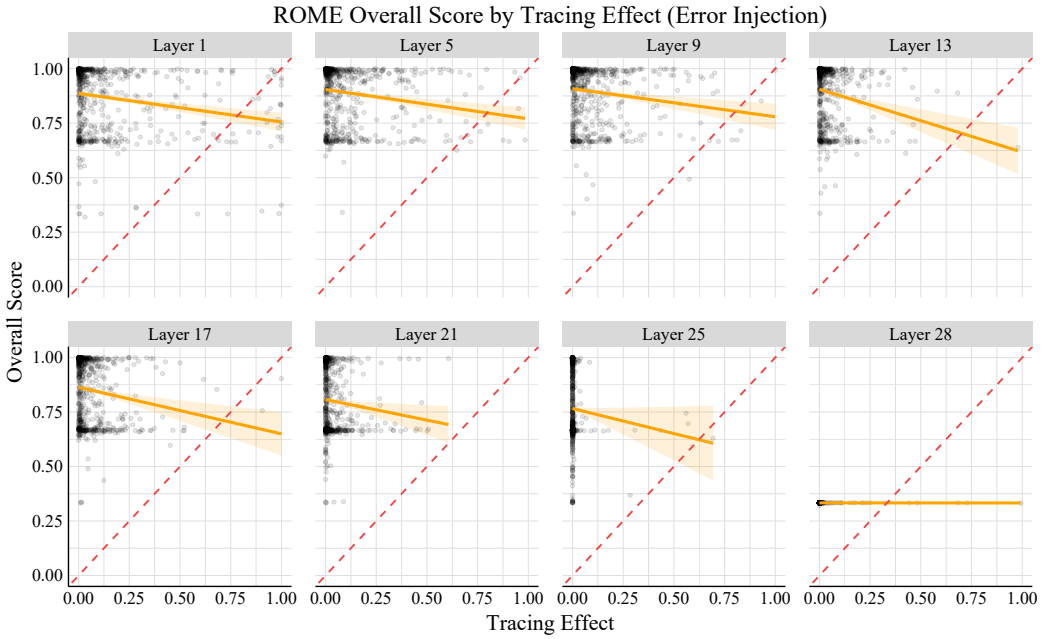


Figure 18: The relationship between ROME **overall score** (average of rewrite/paraphrase/neighborhood scores) and the tracing effect is somewhat negative for most edit layers in the model (for the standard Error Injection editing problem). Red lines show a perfect relationship between tracing effect and edit success, so a negative relationship suggests that tracing localization results do not indicate that editing will be successful.

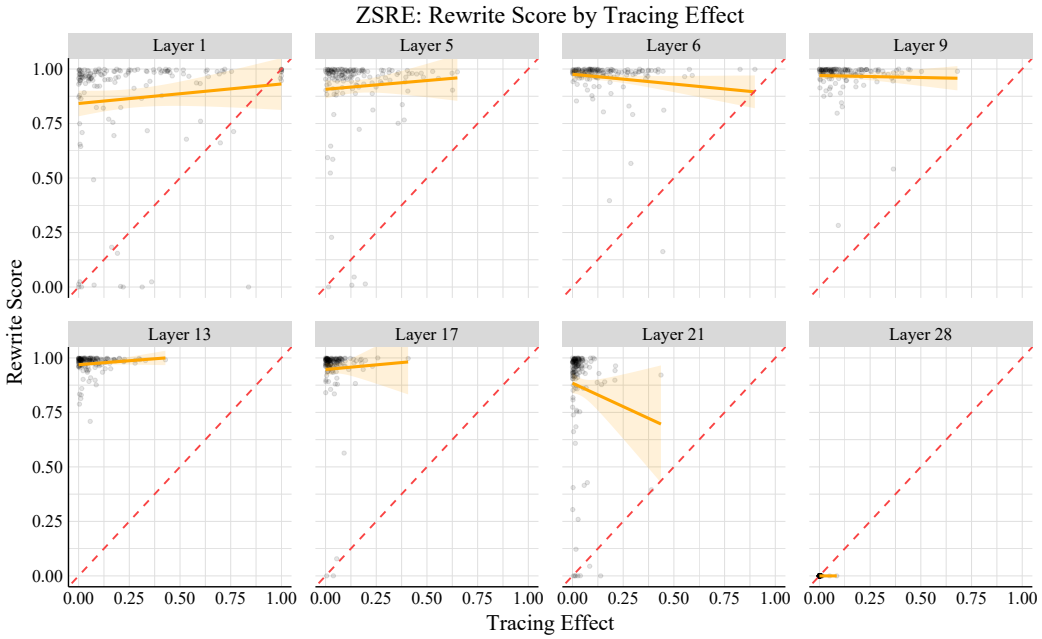


Figure 19: Additional experiments on the **ZSRE** dataset show the same results as for CounterFact, using the ROME editing method with rewrite score as our editing success metric (see regression analysis results in Table 8). Red lines show a perfect relationship between tracing effect and edit success, so near-zero relationships suggest that tracing localization results do not indicate that editing will be successful.